

Project Lingua ex machina / Language Machine (Draft)

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Abstract:

This evolving early stage project aims to develop an offline, open-source generative AI platform to support under-represented languages within the humanities. Designed to run on ordinary consumer hardware without cloud services or complex software infrastructure, the system will enable search, textual analysis, and generative workflows over locally hosted corpora while preserving scholarly control over data.

Contemporary AI systems largely exclude historical and low-resource languages, forcing scholars to rely on commercial platforms optimised for modern, high-resource vernaculars. This creates structural barriers for classicists, educators, and cultural institutions that require stable, transparent, and locally controlled research tools. Our project addresses this gap by aligning language technology with humanistic research practices and European priorities of linguistic diversity and digital sovereignty.

Latin serves as the primary testbed, not as a legacy language but as a linguistically demanding case study. Its rich morphology, complex syntax, and well-structured textual traditions allow us to evaluate NLP methods under conditions that resist simplistic statistical modelling.

Our design addresses, beyond computational constraints, several well-documented limitations of contemporary retrieval-augmented generation (RAG) models, including semantic collapse, the statistical limits of dense retrieval at scale, and misaligned chunking strategies. Central to our approach is a focus on textual structure and variable retrieval granularity, enabling AI-assisted inquiry that respects the organisation, semantics, and interpretive conventions of Latin texts, rather than reducing them to arbitrary fragments.

Keywords:

Digital Latin; Digital Humanities; Hybrid Search; sBERT Embeddings; GGML; Dense Vector Retrieval; Retrieval-Augmented Generation; Semantic Search; Textual Structure; Privacy-Preserving AI; Local Corpora; Textual Structure; Variable Retrieval Granularity; Philology; Digital Sovereignty; Computationally Constrained; Low-Energy AI.

More detailed description:

This project sets out to develop an offline, open-source generative AI platform designed to run on standard consumer hardware and to be adaptable to any language. It will integrate neural code generation with word-based (lexical) and semantic (vector) search over curated, locally hosted corpora, thereby enabling search, document analysis, and generative text workflows without reliance on external servers or specialised infrastructure. The system is designed to require neither Docker nor Python, and avoid complex software dependencies. The code-base is primarily C++.

The primary objective is to deliver an accessible, privacy-respecting solution for underserved and minority languages across Europe. By doing so, the project contributes to linguistic diversity, digital

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Project Lingua ex machina / Language Machine (Draft)

sovereignty, and equitable access to advanced language technologies. As an initial testbed, we focus on Latin and Ancient Greek.

The majority of contemporary AI systems support only a small set of high-resource languages and typically require users to upload data to remote cloud services. Such architectures create structural barriers for minority language communities, cultural heritage institutions, educators, and public-sector organisations that require secure, long-term, locally controlled tools rather than commercial platforms built around data extraction and centralised infrastructure.

Recognising that most users are not trained information specialists, our system employs generative AI to construct sophisticated hybrid searches directly from natural-language queries. Latin and Ancient Greek are not treated as final targets, but as demanding test cases. They provide extensive open-access corpora, exhibit highly complex grammatical structures that place significant stress on NLP systems, and are supported by active digital humanities communities capable of offering informed and structured feedback. These properties make them well suited for validating our approach prior to scaling to genuinely underserved and endangered European languages.

The project directly addresses the needs of users who prioritise privacy and refuse to outsource sensitive data. It aligns closely with European priorities concerning digital sovereignty, linguistic diversity, and the protection of shared knowledge commons, ensuring that modern language technologies remain accessible without compromising user control or cultural value.

At the core of the system lies the re-Issearch engine, extended to support dense vector search. Its successor project, *Schmate*, is implemented in C++ and leverages the GGML tensor library via *llama.cpp* and *bert.cpp* to enable efficient on-device machine-learning inference. The emphasis on high-performance tensor operations on commodity hardware is central to the project’s design philosophy. Our support of GGUF (GPT-Generated Unified Format) for LLMs opens the doors to many thousands of models from platforms like HuggingFace. While other formats exist (like Safetensors for GPU/general use or raw PyTorch) most of these can be converted to GGUF.

Semantic search is driven by sBERT embeddings, while dense vector retrieval is implemented using the HNSW algorithm, with architectural provisions for future support of additional approximate nearest-neighbour models. The widely adopted HNSWlib has been substantially extended to incorporate multiple quantisation methods—including Ternary, RaBitQ³, and Extended RaBitQ (Gao, You et al.). We also implemented re-scoring to also, on demand, use an optimised disk storage alongside LSM-tree-based vector storage, reranking mechanisms, hyperparameter optimisation, and advanced SIMD implementations supporting NEON and SVE. These SIMD optimisations yield performance improvements of approximately four to six times on ARM architectures. GGML itself supports a broad range of GPUs, NPUs, and SIMD-capable processors.

With the observation that many models are pre-quantised (such as the all-MiniLM-L6-v2 with 4-bit quantization which is only 14MB) we also implemented a “pass-through” to enable native support of 1,2,3,4,5,6,8 and 16-bit integer and Fp16 pre-quantised models. Embeddings from a 4-bit model are, for example, handled as a 4-bit ints and not 32-bit floats. This is particularly relevant to GGUF models as they are most often quantized.

³ <https://arxiv.org/abs/2405.12497>; RaBitQ: Quantizing High-Dimensional Vectors with a Theoretical Error Bound for Approximate Nearest Neighbor Search; Geo & Long, 2024

Project Lingua ex machina / Language Machine (Draft)

The lexical component of the hybrid search architecture is likewise highly optimised and differs fundamentally from conventional inverted-index approaches. It imposes no practical limits on the number of fields, paths, word lengths, or term frequencies, and conceptualises documents as graph structures rather than flat records.

This architecture naturally supports multiple object types beyond lexical text and vectors, including numerical values, ranges, dates, phonetic encodings, and various hashing schemes. Such capabilities are particularly valuable for temporal queries, as they enable the retrieval of dates expressed in diverse formats or even alternative calendrical systems, provided these can be internally parsed and normalised.

A further departure from conventional search systems lies in the project's approach to retrieval granularity. Traditional text indexers treat the document or record as both the unit of indexing and retrieval. In contrast, our system supports a dynamic, query-time unit of retrieval that may be specified by the user or inferred heuristically. Document structure is explicitly exploited to identify the most relevant components—such as chapters, pages, paragraphs, or lines—allowing retrieval at sub-document or supra-document levels as appropriate.

This design addresses several well-documented limitations of contemporary retrieval-augmented generation (RAG) models, including semantic collapse, the statistical limits of dense retrieval at scale, and misaligned chunking strategies. As corpora grow, embedding distances tend to compress, clusters overlap, and similarity metrics become less reliable indicators of relevance. Rather than treating text as arbitrarily segmented chunks, our approach models text as structured discourse. Since texts are written to be read, they adhere to conventions that convey meaning through syntax and organisation, which in turn provide a grounding for semantics and contextual interpretation.

For RAG-based queries, we draw on curated, multi-layer corpora of Latin and Ancient Greek, most notably *Opera Latina Adnotata* and *Opera Graeca Adnotata*. These resources are further supplemented and refined through domain expertise, drawing on Adam's over eight years of professional experience in Latin lexicography.

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Project Lingua ex machina / Language Machine (Draft)

History:

Language Machine builds on Schmate. Schmate builds on re-Isearch. re-Isearch itself builds on previous projects: IB and Isite/Isearch. In this sense it is a continuation of over 30 years of R&D.

Isearch was an open-source text retrieval software first developed in 1994 as part of the Isite Z39.50 information framework. The project started at the Clearinghouse for Networked Information Discovery and Retrieval (CNIDR) of the North Carolina supercomputing center MCNC and funded by the National Science Foundation to follow in the track of WAIS⁴ (Wide Area Information Server) and develop prototype systems for distributed information networks encompassing Internet applications, library catalogs and other information resources. Edward Zimmermann having worked both on WAIS and his own Z39.50 server, seeing the need for a massive redesign, quickly joined the fray with his company BSn. From 1994 to 1998 most of the development was centered on the Clearinghouse for Networked Information Discovery and Retrieval (CNIDR) in Research Triangle, North Carolina and BSn in Munich. By 1998 much of the open-source core developers re-focused development into several spin-offs. A fork went on to power search for the US Patents and Trademark office. In 1998 it also became part of the Advanced Search Facility reference software platform funded by the U.S. Department of Commerce. Others joined a team at WIPO in Geneva. Another fork powered the Open Directory Project. One of the main split-offs was the (closed source and proprietary) IB engine developed by BSn. With some new algorithms that solved some of the performance issues with the Pat Arrays it was bespoke deployed in a large number of high profile public projects and collaborations (on 6 continents).

IB also formed the heart of a number of BSn's public projects supported by: EU DG, German Federal Ministry for Education, Family Affairs, Senior Citizens and Youth; German Institute for Political Education; European Union Youth Program; Rosa Luxemburg Foundation; and other orgs.

Development of IB halted in 2011 as its main developers moved on to other projects. It was re-born in 2020 during the COVID pandemic as re-Isearch under an Apache 2.0 license.

⁴ A distributed information server started in 1989 at Thinking Machine Corp by Brewster Kahle currently of the Internet Archive. Inspired by the WAIS project on full-text databases and emerging SGML projects, Z39.50 version 2 (Z39.50:1992) was released.